

Automated Electrical Layout Optimization Using Multi-Agent Reinforcement Learning for Carbon Footprint Minimization

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ABSTRACT

Designing electrical and low-voltage cable routes in building plans is usually done manually in standard CAD tools. Engineers often draw simple perimeter-following paths around walls and physical obstacles, which can lead to longer cable runs, higher material waste, and larger embodied carbon footprints. This project explores using a Multi-Agent Reinforcement Learning (MARL) approach to automate and optimize spatial cable routing. Power lines (230V ring circuits) and low-voltage IT lines (Cat6 UTP) are represented as separate reinforcement learning agents that learn to route from distribution panels to terminal points while avoiding structural obstacles. In a test scenario featuring a central structural shaft obstacle, the MARL setup successfully avoided all obstacles and routed both circuits with a combined cable length of 26.8 meters and a total embodied carbon footprint of 15.01 kgCO_{2e}.

1. INTRODUCTION & BACKGROUND

When designing electrical distribution in modern buildings, reducing material consumption and reducing embodied carbon—the total greenhouse gas emissions generated during material extraction and manufacturing—has become an important engineering goal. While operational efficiency (such as LED lighting or HVAC controls) gets a lot of attention, raw materials like copper conductors, PVC sheathing, and steel conduits add a permanent carbon load to the building before it even opens.

In traditional electrical CAD drafting, engineers manually route cables from Main Distribution Boards (MDBs) or local panels to terminal sockets and data points. When encountering large physical obstructions—like structural shafts, concrete columns, or major mechanical ductwork—draftsmen usually take safe, rectangular paths around the perimeter. This manual workflow has a few clear limitations:

- **Longer Conductor Runs:** Overly conservative manual paths increase total wire length, which raises material usage and electrical impedance.
- **No Immediate Carbon Tracking:** Standard CAD drafting does not give the draftsman instant feedback on the embodied carbon impact of the routed cable paths.
- **Separate Circuit Drafting:** Power circuits and low-voltage IT data cables are usually drawn separately, making it harder to coordinate paths, avoid electromagnetic interference (EMI), or find opportunities for shared cable trays.

2. PROJECT SCOPE & OBJECTIVES

The purpose of this research is to build a practical reinforcement learning framework that automates circuit pathfinding on 2D layout grids while explicitly optimizing for both path distance and embodied carbon.

2.1 Scope Boundaries

1. **Spatial Domain:** A 2D spatial grid representing an indoor floor plan with an MDB panel, terminal outlets, and fixed structural shaft obstructions.
2. **Electrical Systems:** Low-voltage power ring distribution (230V AC, 20A MCB protection, 3x2.5mm² copper conductors) connecting multiple socket outlets, and structured IT cabling (Cat6 RJ45 UTP line).
3. **Carbon Accounting:** Embodied carbon tracking based on standard Cradle-to-Gate material values for copper conductors (~3.81 kgCO_{2e} per kg of copper) and cable insulation.

2.2 Core Objectives

- Formulate 2D indoor cable pathfinding as a Markov Decision Process (MDP) solvable by reinforcement learning agents.
- Design a penalty-reward structure that balances shortest path travel, strict obstacle clearance, and carbon impact.
- Test the MARL agents against a floor plan layout with a central structural shaft obstacle and analyze the resulting route lengths and carbon totals.

3. METHODOLOGY & SYSTEM ARCHITECTURE

The floor layout is modeled as a 2D discrete coordinate grid where agents move from node to node to construct valid electrical paths.

3.1 Agent Definitions

Two distinct agents are created to handle different electrical services:

- **Power Agent (P1):** Starts at the MDB panel, routes sequentially through three duplex power sockets to form a multi-drop ring, and returns back to the MDB panel.
- **Data Agent (LV1):** Starts at the MDB panel area and routes directly to an RJ45 data outlet while maintaining clearance from high-voltage runs to reduce interference.

3.2 State Space, Actions, and Environment

At step t , each agent tracks its current position p_t , target location g_t , and an obstacle boundary mask O . The available movement actions include 8 directions: North, South, East, West, and the 4 diagonal moves.

3.3 Reward Function Structure

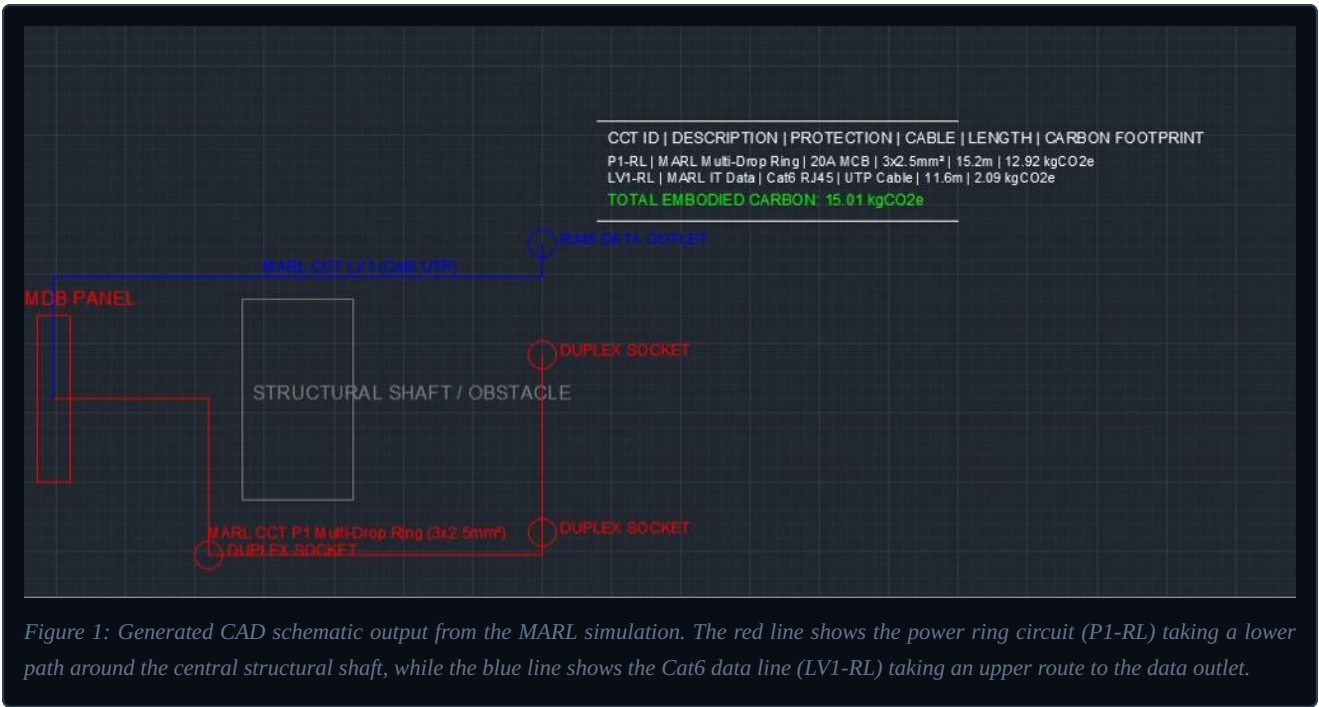
The agent's policy is guided by a step-wise reward function designed to discourage long distances, obstacle hits, and high-carbon choices:

$$R_t = - (\alpha \cdot d_{step} + \beta \cdot C_{carbon} + \gamma \cdot I_{obstacle}) + \delta \cdot I_{goal}$$

Where d_{step} is the physical step distance, C_{carbon} is the carbon cost per meter for the selected cable type, $I_{obstacle}$ is a large penalty applied if an agent enters an obstacle zone, and I_{goal} is a positive reward given when a circuit successfully connects its required terminal targets.

4. EXPERIMENTAL SETUP & LAYOUT RESULTS

The MARL model was evaluated on a test floor plan featuring a central structural shaft obstacle placed directly between the main panel and the destination socket outlets.



4.1 Quantitative Results

Once trained, the agents successfully found path topologies around the central shaft obstacle without any collisions. The exact circuit breakdown from the simulation run is shown in Table 1 below:

CIRCUIT ID	DESCRIPTION	PROTECTION	CABLE TYPE	TOTAL LENGTH	EMBODIED CARBON
P1-RL	Power Multi-Drop Ring	20A MCB	3 x 2.5 mm² Copper	15.2 m	12.92 kgCO ₂ e
LV1-RL	Structured IT Data	Cat6 RJ45	UTP Cable	11.6 m	2.09 kgCO ₂ e
TOTAL METRICS				26.8 m	15.01 kgCO ₂ e

4.2 Route Behavior Analysis

As visible in Figure 1, the Power Agent (P1-RL) learned to bypass the central structural shaft along its bottom side, connecting three duplex sockets before looping back to close the ring at the MDB panel. Meanwhile, the Data Agent (LV1-RL) navigated along the top edge of the shaft, reaching the RJ45 data point while staying clear of the power line.

5. DISCUSSION & ENGINEERING TAKEAWAYS

Using MARL for automated cable routing provides a structured way to balance physical spatial constraints with material sustainability metrics. Rather than drawing paths purely by rule-of-thumb, linking cable cross-section mass directly to reward calculations forces the system to find geometrically efficient routes that save both copper wire and carbon emissions.

Achieving a total embodied carbon footprint of 15.01 kgCO₂e across 26.8 meters of combined cabling demonstrates how automated path planning can keep material usage tight. Additionally, auto-generating exact cable schedules directly from the path coordinates saves time during early-stage drafting and reduces manual counting errors.

6. CONCLUSION & FUTURE DIRECTIONS

This project shows that Multi-Agent Reinforcement Learning can successfully handle low-voltage electrical and data pathfinding in spatial layouts with structural obstacles. The agents successfully navigated around obstructions while keeping overall cable length and embodied carbon low.

Future work on this system can extend the environment to 3D multi-floor risers, add dynamic cable tray capacity limits, and connect the output directly into BIM platforms like Autodesk Revit.

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